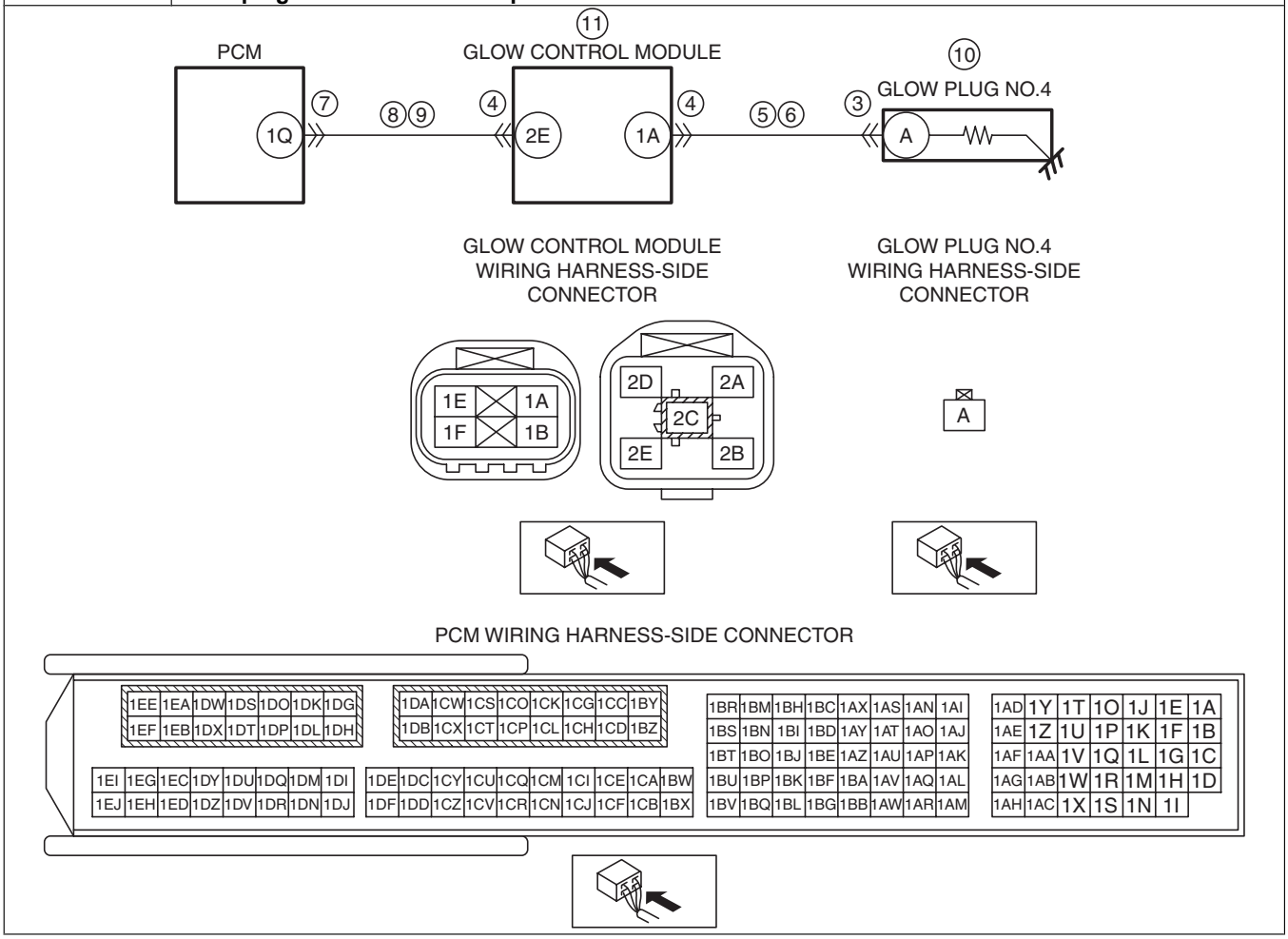
DTC P0674:00 [SKYACTIV-D 1.5]

id0102q2213600

DTC P0674:00	Glow plug No.4 control circuit problem
DETECTION CONDITION	• When the following conditions are met, a condition continues for **5 s** in which the current at the glow plug No. 4 circuit is **less than 2.88 A** while glow control is being executed at a control voltage of **75% or more** of the battery voltage, or the at the glow plug No.4 circuit **exceeds 65 A** while glow control is being executed at a control voltage of **less than 75%** of the battery voltage. **MONITORING CONDITIONS** — 0.5 s have elapsed after ignition was switched ON. (engine off or on) — During glow control — Glow plug No.1 contol circuit voltage: **3—11.3 V** — Battery voltage: **8—20 V** — The following DTC is not detected: • LIN communication system: U0106:00 **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Not applicable
POSSIBLE CAUSE	• Glow plug No.4 connector or terminals malfunction • Glow control module connector or terminals malfunction • Open circuit in wiring harness between glow plug No.4 terminal A and glow control module terminal 1A • Short to ground in wiring harness between glow plug No.4 terminal A and glow control module terminal 1A • Short to power supply in wiring harness between glow plug No.4 terminal A and glow control module terminal 1A • PCM connector or terminals malfunction • Short to power supply in wiring harness between glow control module terminal 2E and PCM terminal 1Q • Open circuit in wiring harness between glow control module terminal 2E and PCM terminal 1Q • Glow plug No.4 malfunction • Glow control module malfunction • PCM malfunction

DTC P0674:00	Glow plug No.4 control circuit problem



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT GLOW PLUG NO.4 CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the glow plug No.4 connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 12.
		No	Go to the next step.
4	**INSPECT GLOW CONTROL MODULE CONNECTOR CONDITION** • Disconnect the glow control module connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 12.
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**INSPECT GLOW PLUG NO.4 CONTROL CIRCUIT FOR OPEN CIRCUIT** • Verify that the glow plug No.4 and glow control module connectors are disconnected. • Inspect for continuity between glow plug No.4 terminal A (wiring harness-side) and glow control module terminal 1A (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between glow plug No.4 terminal A and glow control module terminal 1A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 12.
6	**INSPECT GLOW PLUG NO.4 CONTROL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the glow plug No.4 and glow control module connectors are disconnected. • Switch the ignition ON (engine off). • Measure the voltage at the glow plug No.4 terminal A (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between glow plug No.4 terminal A and glow control module terminal 1A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 12.
7	**INSPECT PCM CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 12.
		No	Go to the next step.
8	**INSPECT GLOW CONTROL MODULE SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the glow plug No.1, glow control module and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the glow control module terminal C (wiring harness-side). • Is the voltage 0 V?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between glow control module terminal 2E and PCM terminal 1Q. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 12.
9	**INSPECT GLOW CONTROL MODULE SIGNAL CIRCUIT FOR OPEN CIRCUIT** • Verify that the glow plug No.1, glow control module and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between glow control module terminal 2E (wiring harness-side) and PCM terminal 1Q (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between glow control module terminal 2E and PCM terminal 1Q. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 12.

STEP	INSPECTION		ACTION
10	**INSPECT GLOW PLUG NO.4** • Inspect the glow plug No.4. (See GLOW PLUG INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the glow plug No.4, then go to Step 12. (See GLOW PLUG REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
11	**INSPECT GLOW CONTROL MODULE** • Inspect the glow control module. (See GLOW PLUG CONTROL MODULE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the glow control module, then go to the next step. (See GLOW PLUG CONTROL MODULE REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
12	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOEO self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
13	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.